

Multiple Choice (10 marks)

1. What type of diet is recommended for weight maintenance?
 - (a) Low protein and low GI
 - (b) High protein and high GI
 - (c) Low protein and high GI
 - (d) High protein and low GI
2. Which of the following vitamins provides the coenzyme for oxidative decarboxylation of pyruvate?
 - (a) Folate
 - (b) Niacin
 - (c) Riboflavin
 - (d) Thiamin
3. Which of the following statements is correct (according to knowledge in 2020)?
 - (a) Consumers with phenylketonuria must avoid the consumption of the sweetener aspartame
 - (b) Consumers with phenylketonuria must avoid the consumption of the sweetener saccharin
 - (c) Consumers with phenylketonuria must avoid the consumption of the sweetener sucralose
 - (d) Consumers with phenylketonuria must avoid the consumption of the sweetener acesulfame K
4. Oral bacterial synthesise extracellular glucans (dextran and mutan) from which host dietary component?
 - (a) glucose
 - (b) Alpha limit dextrans
 - (c) Sucrose
 - (d) polysaccharides
5. Which vitamins are important in lowering circulating homocysteine levels
 - (a) Vitamin D
 - (b) Vitamin C
 - (c) Vitamin A
 - (d) Folate, vitamins B 6 and B 12

True / False (10 marks)

Answer True or False

6. True or False: Rising incomes and urbanization globally lead to an increased demand for meat and dairy products.
7. True or False: People with this disorder typically have normal energy intake, normal body weight, and hyperthyroidism.
8. True or False: The basal metabolic rate refers to the body's adjustment of heat production to changes in nutrition and energy balance.
9. True or False: A deficiency of thiamin in alcoholics can cause Cushing syndrome, which is related to cortisol levels.
10. True or False: Plasma lipoproteins have a hydrophobic core of triacylglycerol and cholesterol esters.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the roles of intestinal "brakes" and "accelerators" in digestion, and analyze why the statement about dietary starch and its effect on small intestinal motility might be misleading.
12. Discuss the role of peroxisomes and mitochondria in fatty acid -oxidation, and analyze why the statement about their functions might be misleading.

End of Paper